

YORDIN HERRERA BORGE

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Professional Profile

Data Science Engineering student (LEAD University, expected 2027) with academic and personal-project experience building and validating classification models, particularly logistic regression (statsmodels/MLE), exploratory data analysis (EDA), hypothesis testing (ANOVA, MANOVA), and dimensionality reduction (PCA, MCA). Complements analytical work with full-stack development (React, Node.js, FastAPI), translating data and models into deployed, working software.

Technical Skills

- **Classification & Predictive Modeling:** Logistic Regression (Logit/MLE), Multiple Linear Regression (OLS), model validation (pseudo R^2 , p-values).
- **Inferential Statistics:** ANOVA, MANOVA, Student's t-test, post-hoc comparisons.
- **Dimensionality Reduction:** PCA, MCA, biplots, scree plots, correlation circles.
- **Clustering:** K-Means, DBSCAN, Elbow Method, Silhouette Score.
- **Languages & Tools:** Python (Pandas, NumPy, scikit-learn, statsmodels, SciPy), SQL, JavaScript/TypeScript.
- **Databases:** PostgreSQL, relational modeling and querying.
- **Visualization:** Matplotlib, Seaborn, Altair, Plotly.
- **Development & Deployment:** FastAPI, Docker, GitHub Actions (CI/CD), Render/Railway.

Data Mining & Statistical Modeling Projects (Academic)

Classification with Logistic Regression — Personality Profiles [GitHub](#)

Python, statsmodels, scikit-learn

- Trained a Logit model (MLE) on 20,000 observations to classify high-leadership profiles from behavioral variables; achieved Pseudo R^2 of 0.22 with significant predictors ($p < 0.001$).
- Cross-validated findings using Student's t-test and MANOVA for multivariate group differences.

Unsupervised Learning — Clustering [GitHub](#)

Python, scikit-learn, K-Means, DBSCAN, Matplotlib

- Applied K-Means and DBSCAN clustering; evaluated cluster quality via Elbow Method and Silhouette Score, comparing sensitivity to outliers and cluster shape.

Dimensionality Reduction — PCA Multidomain [GitHub](#)

Python, scikit-learn, NumPy, Altair

- Applied PCA across three domains (automotive market, socioeconomic indicators for 167 countries, Telco customer behavior); built scree plots, correlation circles, and biplots to identify the optimal components.
- Projected customers onto principal components to profile segments with higher churn risk.

Multiple Correspondence Analysis (MCA) — Categorical Variables [GitHub](#)

Python, prince

- Applied MCA to qualitative datasets (commercial flights, banking data, Telco customers) where PCA does not apply; ran comparative PCA vs. MCA evaluating explained inertia and $\cos^2$.

Multiple Linear Regression & ANOVA — Nutritional Analysis [GitHub](#)

Python, statsmodels, SciPy

- Built an OLS model to predict caloric density from macronutrient composition; ran one-way ANOVA and post-hoc tests on sugar-content differences across food categories.

Software Engineering Projects

FactualQ — Billing Analytics Dashboard [GitHub](#)

[Live demo](#)

React, TypeScript, Node.js, PostgreSQL, Docker, JWT, GitHub Actions

- Full-stack BI platform with CSV ingestion, KPI dashboards, JWT authentication, and dynamic data normalization; deployed on Railway/Render with a modular architecture.

Background Remover — Computer Vision App [Frontend](#)

[Backend](#) · [Live demo](#)

Python, ONNX Runtime, rembg, Streamlit

- ML web app for automated background removal using U2Net (ONNX); built a layered REST backend plus a Streamlit local version with in-memory inference.

Hand Gesture Detection [GitHub](#)

Python, OpenCV, MediaPipe, NumPy

- Real-time hand landmark detection on live video; optimized the processing pipeline with NumPy for real-time performance.

Work Experience

IT Support Technician — Librería Cristal

Jan 2022 – Present

- Tier 1/2 support for 15+ workstations (~20% downtime reduction); installed network infrastructure and documented 10+ troubleshooting workflows.

Education

B.S. Data Science Engineering — LEAD University, Costa Rica

2025 – 2027

Data Mining · Statistical Inference · Machine Learning · Databases · Software Engineering

Technical Degree in Software Development — CTP Francisco J. Orlich

2020 – 2022

Certifications & Languages

INA: Software Quality Control · Spreadsheet Development · Basic English · Intermediate English

Spanish: Native **English:** B1 (Intermediate), currently progressing toward B2